

Grid Synchronization Phase-Locked Loop Strategy for Unbalance and Harmonic Distortion Conditions

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Abstract The synthesis, design and analysis of a three-phase phase-locked loop (PLL) algorithm under grid voltage uncertainties is presented. Unlike other techniques, the proposed strategy is simple but yet, robust against unbalanced and distorted voltage conditions. The method does not rely on the symmetry of the three-phase voltages, like conventional PLL implementation techniques. Under phase lock, the reference voltage and the output voltage are in synchronism, hence they cross the zero axis at the same time. The proposed PLL implementation exploits this fact, by counting the difference between the zero crossing times of the grid voltage and the output voltage waveform, and generating an error signal proportional to this difference. This error is fed into a PI regulator which forces it to zero over time. The proposed algorithm is compared against two popular PLL techniques. Simulation and experimental results show better performance of the proposed PLL implementation in unbalanced grid voltage conditions.

Keywords Phase-locked loop · Grid voltage distortion · Power electronics converters · Phase detection · Grid synchronization

1 Introduction

Renewable energy has, in the last decade, emerged as a viable tool for combating climate change, hence its widespread popularity in industry and academia. It is estimated that a third of global electricity generation will be supplied by renewable energy by 2035 (State Grid Corporation of China 2012; Hammons et al. 2000). A comprehensive survey of renewable energy systems, as they pertain to sustainable energy, was undertaken by Nehrir et al. (2011). New trends in power electronic technology are examined by Carrasco et al. (2006) for interfacing these renewable energy sources with the grid to maintain power supply reliability. The wind turbine is one of the many forms of renewable energy generator that has experienced rapid growth. However, interconnecting such a renewable energy system to the power grid requires a robust phase-locked loop (PLL) algorithm capable of dealing with uncertainties. In light of this, alternative synchronization strategies are necessary to fully exploit this energy source and keep pace with ever-growing energy needs.

A PLL is a closed-loop control system that modulates the phase angle of an output signal, so that it closely matches the phase angle of a reference input signal. It essentially minimizes the phase error between both signals, thus matching their oscillatory frequencies. The general technique of doing this is generating a voltage signal corresponding to the phase difference/frequency difference between the output signal and the reference signal. Thus, if the input signal is frequency- or phase-modulated, the output becomes the demodulated output of the PLL (Phipps et al. 2006).

A PLL implementation can be linear (S-domain), digital, or software. The general structure of any PLL consists of a phase detector (PD), a loop filter (LF), and a voltage controlled oscillator (VCO). The phase detector compares the phase angle of the reference signal, usually a

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continuous-time signal, with the phase angle of the VCO output signal. The error signal generated by the phase detection process is filtered by the low pass filter, as we are only interested in the dc varying components. The filtered voltage controls an oscillator to produce a feedback signal with adjusted frequency and phase, such that its phase angle locks into that of the reference signal. In general, it can be stated that in designing a phase-locked loop for most purposes, a trade-off has to be made between achieving high system stability at the expense of system performance or vice-versa. By using

A practical three phase power system is far from ideal and involves various single-phase loads distributed asymmetrically between the feeders, loads constantly being connected and disconnected, and nonlinear loads that distort the voltage of the grid at the point of common coupling (PCC). Additionally, the reference signal obtained from the grid for grid-tie inverters may be distorted by harmonics, which may be local to the power converter itself or generated at some other point on the grid. It is usual to find nonlinear loads operate in an asymmetrical manner; hence, these harmonics may not always be evenly distributed among the three-phase. This uneven distribution of harmonics causes distortions to the grid voltage. Another causative factor of asymmetrically distributed harmonics on the grid is the connection of nonlinear single-phase loads (Wang and Li 2010). Voltage offsets are another form of waveform distortion introduced into three-phase system often due to failure of grid-connected rectifiers (Seymour 2005). Passos et al. (2015) also cites voltage sags as a common disturbance in the power grid at the point of common coupling. Under these conditions, the frequency and phase of the grid is uncertain and difficult to determine, but remains crucial for converter operation and control. The past couple of years have seen various implementations of the PLL with varying adaptations to make them more responsive to changing grid conditions and a comparative study of some of them have been carried out by Gao and Barnes (2012).

The design, analysis and optimization of a PLL, which used the d-q transform as a phase detector (PD), for a three-phase utility interface inverter was carried out in Chung (2000). Particularly, the errors due to distorted voltages were analyzed. The use of coordinate transformation to implement a phase detection scheme is popular in multiphase systems, with the synchronous reference frame (SRF) enjoying the most widespread use (Karimi-Ghartemani 2012; Liccardo et al. 2010; Perez et al. 2008; Rodriguez et al. 2007). However, in Phipps et al. (2006), a novel PLL design was proposed which, unlike traditional three-phase PLL, required no coordinate transformation. An optimized digital implementation of PLLs is presented in Freijedo et al. (2011), where the digital oscillator is used to execute trigonometric functions. Jain et al. (2014) focuses on optimizing the dynamic response of the PLL for fast tracking under disturbance. Park

et al. (2014) deal with grid voltage distortion by constructing an observer to extract the positive sequence voltage and hence enhance the performance of the PLL. Most of the aforementioned techniques are complex in their circuit implementation due to the numerous subsystems deployed, while others are far from robust and make impractical assumptions about grid voltage conditions.

This paper proposes a PLL strategy for renewable energy systems. Phase detection is carried out by counting the time difference between the zero crossings of the reference signal and the output signal. While both signals are out of phase, (cross the zero axis at different times), their amplitudes sampled at any point in time would be unequal. An error signal corresponding to this difference in amplitudes is fed to a PI regulator which forces the error to zero over time. This methodology is as simple as it is robust. It does not rely on the symmetry between the amplitudes of the three-phase voltages, so even under conditions of voltage imbalance, it is still able to track the reference frequency. In a similar way, the presence of distortions in the grid voltage does not impede the performance of the PLL system as it would be seen in the simulation and experimental results. The performance of this approach is compared against two classical schemes. The layout of the rest of the paper is as follows: Two popular PLL schemes are briefly introduced in Sect. 2. For simplicity and ease of reference, the proposed PLL strategy would be referred to as “PLL Counter,” the strategy implemented in Chung (2000) is labeled “Synchronous Frame PLL (SRF-PLL),” and the scheme presented in Phipps et al. (2006) is labeled “PW PLL.” The synthesis, operational principle and modeling of the proposed PLL technique is described in Sect. 3. In Sect. 4, simulation results of the three implementations are reported and discussed. Section 5 presents the experimental results of the three different PLL strategies and the conclusion follows.

2 Conventional PLL Schemes

2.1 Synchronous Reference Frame (SRF) PLL

This PLL design technique employs the Direct Clarke and Givens rotating matrix to determine the frequency and phase difference between the utility grid voltage and the PLL output voltage. This method is common in three-phase power systems and the block diagram is shown above in (Fig. 1).

Assuming balanced conditions in the utility voltage, the three-phase utility voltage can be represented as:

$$v_{abc} = V_m \cdot \begin{pmatrix} \cos \theta \\ \cos \left(\theta - \frac{2\pi}{3} \right) \\ \cos \left(\theta + \frac{2\pi}{3} \right) \end{pmatrix} \quad (1)$$

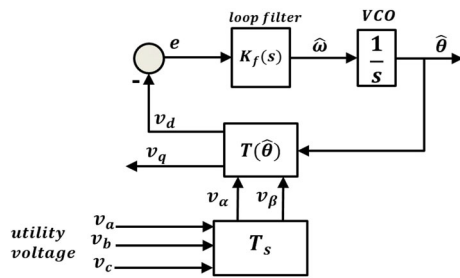


Fig. 1 Block diagram of an SRF-PLL

where $v_{abc} = [v_a \ v_b \ v_c]^T$. Since we are assuming balanced conditions, the utility voltage can be transformed using the Direct Clarke transform in the stationary reference frame as:

$$v_{\alpha\beta} = T_s \cdot v_{abc} \quad (2)$$

where $v_{\alpha\beta} = [v_\alpha \ v_\beta]^T$ and T_s is the Direct Clarke transformation matrix given by:

$$T_s = \frac{2}{3} \cdot \begin{pmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & -\frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} \quad (3)$$

Using the a rotating matrix, these can be transformed into the synchronous reference frame using the PLL output angle $\hat{\theta}$ as:

$$v_{qd} = T(\hat{\theta}) \cdot v_{\alpha\beta} \quad (4)$$

where $v_{qd} = [v_q \ v_d]^T$ and $T(\hat{\theta})$ is the Givens rotation matrix that transforms the voltages into the synchronously rotating reference frame. It is given as:

$$T(\hat{\theta}) = \begin{pmatrix} \cos \hat{\theta} & -\sin \hat{\theta} \\ \sin \hat{\theta} & \cos \hat{\theta} \end{pmatrix} \quad (5)$$

Performing the above transformation, the direct component of the voltage v_d is gotten to be as proven by Chung (2000):

$$v_d = V_m(\sin(\hat{\theta} - \theta)) \quad (6)$$

Setting $E_m = -V_m$ and $\delta = \theta - \hat{\theta}$, we get:

$$v_d = E_m \sin \delta = e \quad (7)$$

Similarly, the angular frequency of the PLL is given by:

$$\hat{\omega} = \frac{d\hat{\theta}}{dt} = K_f e \quad (8)$$

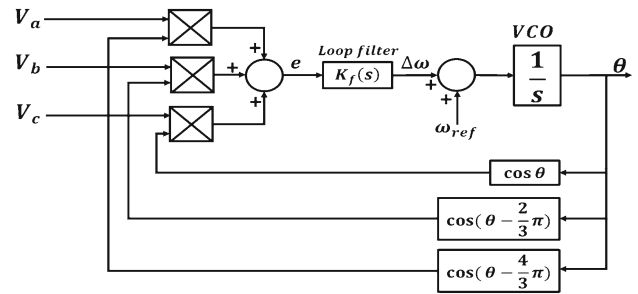


Fig. 2 Block diagram of PLL structure

where K_f is the transfer function of the loop filter, which could be a gain. With the assumption of a very small phase difference between the grid voltage and the PLL output voltage is very small, the equation can be linearized as:

$$e \cong E_m \delta \quad (9)$$

Considering the last two equations, we see that the frequency of the PLL output voltage $\hat{\omega}$ and the phase angle $\hat{\theta}$ can be made to track the frequency of the grid voltage ω and the phase angle θ , respectively, by an appropriate selection of the loop filter gain.

2.2 PW PLL

This PLL system was developed and implemented in Phipps et al. (2006). Quite in contrast to the SRF-PLL, this technique employs no coordinate transformation for the detection of phase difference between the PLL voltage and the grid voltage. The phase detecting mechanism here employs three multipliers, one per phase, that take the product of the grid voltage and the PLL output voltage. These products are then summed together to result in a non-time-varying error signal. The error signal is eliminated by locking the phase angle of the feedback signals into that of the grid voltage. The block diagram of the PLL system is seen in Fig. 2.

After sampling by the DSP, the per-unit values of the three-phase input voltages are given as:

$$\begin{aligned} V_a(k) &= 1 \text{ p.u.} \cdot \sin(\theta(k)) \\ V_b(k) &= 1 \text{ p.u.} \cdot \sin\left(\theta(k) + \frac{2\pi}{3}\right) \\ V_c(k) &= 1 \text{ p.u.} \cdot \sin\left(\theta(k) - \frac{2\pi}{3}\right) \end{aligned} \quad (10)$$

As explained above, these voltages are multiplied with the corresponding PLL output voltages and summed to yield an error signal. The proportional-integral controller of the PLL system uses this error to estimate the difference in angular frequency of the mains voltage $\Delta\omega(k)$ given by:

$$\Delta\omega(k) = \left(K_p + \frac{K_i z}{z-1} \right) e(k) \quad (11)$$

where $\Delta\omega(k)$ is the output of the controller. Next, the reference angular frequency $\omega_{\text{ref}} = 2\pi f$ ($f = 60$ Hz) is added to $\Delta\omega(k)$:

$$\omega(k) = \Delta\omega(k) + \omega_{\text{ref}} \quad (12)$$

The incremental change of the phase angle $\Delta\theta(k)$ is then determined by multiplying $\omega(k)$ by the sampling time, z^{-1} :

$$\Delta\theta(k) = z^{-1}\omega(k) \quad (13)$$

Taking the discrete-time integral of the incremental phase angle $\Delta\theta(k)$ over a full period $[0 - 2\pi]$ would give the estimation of phase angle $\theta(k)$ of the grid voltage:

$$\theta(k) = \frac{z\Delta\theta(k)}{z-1} \quad (14)$$

This estimated angle is then fed back into the phase detectors, where the whole cycle is repeated until the PI drives the phase error to zero, attaining phase lock between the grid voltage and the PLL voltage.

3 Proposed PLL Approach

3.1 Synthesis Method

The proposed PLL does not involve averaging techniques or coordinate transformations for phase difference detection between the phase angle of the grid voltage and the phase angle of PLL voltage, as in the two schemes just examined. Figure 3 details the main blocks deployed in its implementation.

A single-phase of the three-phase grid voltage and the PLL output are both fed into the counter which comprises a zero crossing detector and a edge-triggered summer. Using appropriate blocks, the PLL output waveform and the input grid voltage waveform are represented by sawtooth waveforms, for easier phase difference detection and control. At regular time intervals (in this work, the negative zero crossing of the grid voltage waveform), the difference in the amplitudes of

the sawtooth waveforms is evaluated by a summer; this difference in amplitude corresponds to the difference between phase angle of the grid voltage and the phase angle of the PLL output voltage. If the grid voltage lags the PLL output voltage, the amplitude of its corresponding waveform would always be higher than that of the PLL output voltage. The converse also holds. Hence, by minimizing this amplitude difference in the waveforms, we can effectively drive the error in their phases to zero, achieving phase lock.

3.2 Operational Principle

The dynamics of the system can be modeled as by Phipps et al. (2006) since both implementations only differ in their methods of phase detection methodology. For clarity, the equations would be restated here with little modifications to the notations as they apply to the counter.

As explained earlier, the counter yields an evaluation of the phase error between the grid voltage and the PLL output voltage. This phase error is fed into the PI regulator, which uses it to calculate the difference in angular frequency $\Delta\hat{\omega}(k)$ between the PLL output and the mains voltage as shown below:

$$\Delta\hat{\omega}(k) = \left(K_p + \frac{K_i z}{z-1} \right) e(k) \quad (15)$$

Next, the reference angular frequency $\omega_{\text{ref}} = 2\pi f$ is added to $\Delta\hat{\omega}(k)$. This is the free-running frequency of VCO (not the waveform amplitude), that it would run at when the error has been driven to zero.

$$\hat{\omega}(k) = \Delta\hat{\omega}(k) + \omega_{\text{ref}} \quad (16)$$

To get the incremental change in the phase angle $\Delta\hat{\theta}(k)$, $\hat{\omega}(k)$ is multiplied by z^{-1} .

$$\Delta\hat{\theta}(k) = z^{-1}\hat{\omega}(k) \quad (17)$$

Taking the discrete-time integral of the incremental phase angle $\Delta\hat{\theta}(k)$ over a full period $[0 - 2\pi]$ would yield an estimation of phase angle $\hat{\theta}(k)$ of the grid voltage:

$$\hat{\theta}(k) = \frac{z\Delta\hat{\theta}(k)}{z-1} \quad (18)$$

This value of the phase angle is then fed back into the counter, and the whole process repeated until the error is driven to zero.

3.3 Modeling

A PLL system is usually constrained to operate within a band of frequencies known as the lock range. Operation of the PLL

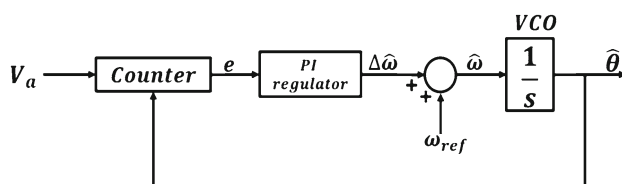


Fig. 3 PLL Counter block diagram

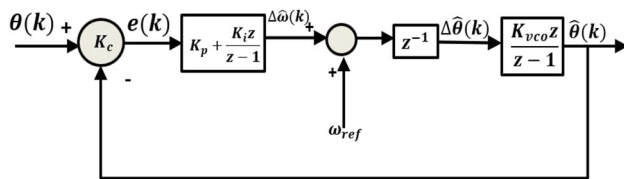


Fig. 4 Linear discrete model of proposed PLL

outside this range results in instability, i.e., phase tracking can no longer be guaranteed (Xiu et al. 2004). Within this lock range, the model of the PLL can be linearized, so that its output can be regulated by a linear controller. The linear model is presented in Fig. 4

Here, K_p and K_i represent the gains of the PI controller. The phase detector output is conditioned by the gain of the counter represented by K_c . K_{vco} denotes the gain of the voltage controlled oscillator. These values were determined empirically. The discrete transfer function can now be written as:

$$\frac{\hat{\theta}(k)}{\theta(k)} = \frac{(\alpha + \beta)z - \alpha}{z^2 + (\alpha + \beta - 1)z + (1 - \alpha)} \quad (19)$$

where $\alpha = K_{vco}K_cK_p$ and $\beta = K_{vco}K_cK_i$.

4 Simulation Results

4.1 Setup

A series of simulations have been carried out using the three different implementations to establish the capability of the proposed technique. The proposed system's parameters are summarized as follows: $K_{vco} = 1$, $K_c = 60$, $K_p = 10$ and $K_i = 33.33$. The parameters for the SF-PLL are given by: $K_p = 20$ and $K_{vco} = 1$. For the PW PLL, the parameters are: $K_p = 15$ and $K_{vco} = 1$. In all cases, the corner frequency of the VCO, ω_{ref} , is set to 60 Hz. The control algorithm is implemented using Matlab/Simulink with the sampling frequency set to 10 kHz.

4.2 Results

The responses of the three different PLL systems under the condition of grid voltage imbalance are seen in Fig. 5. In Fig. 5b, the error of the synchronous reference frame (SRF-PLL) under phase imbalance is shown. The voltage of phase A is kept at the nominal value of the utility voltage, while phase B is allowed to sag to a level of 90 % and phase C is allowed to spike to 110 % of the nominal voltage at 0.5 s. It can be seen that at that instant, the phase error stops being zero and starts oscillating instead. The PW PLL is subjected

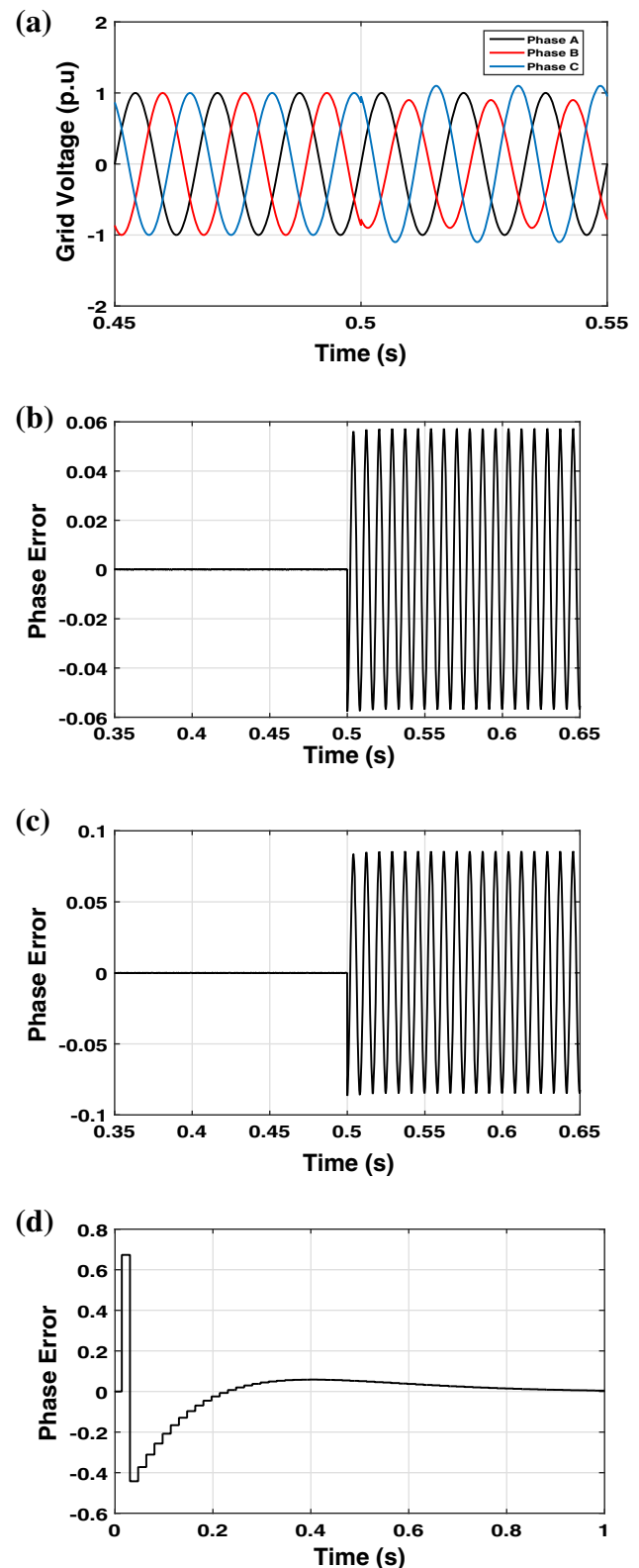


Fig. 5 Response to grid voltage imbalance of: **a** grid voltage under phase imbalance conditions. **b** SRF-PLL. **c** Summer (PW PLL). **d** PLL Counter

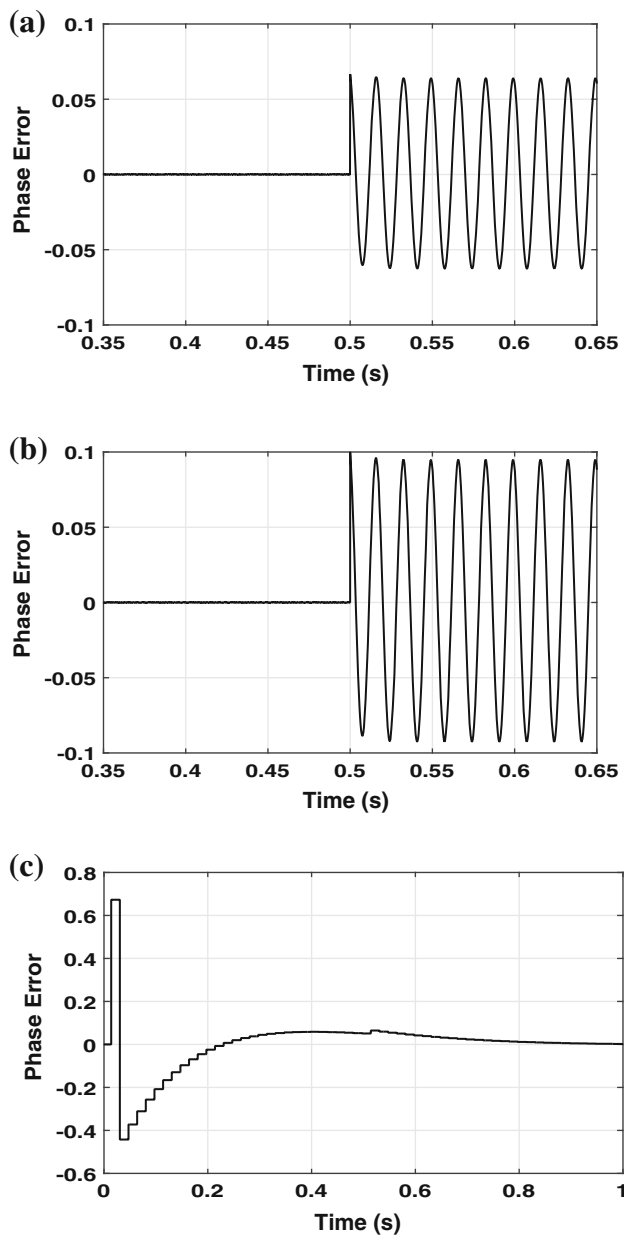


Fig. 6 Response to voltage offset of different PLL techniques. **a** Synchronous reference PLL. **b** PW PLL. **c** PLL Counter

to the same condition of phase imbalance in Fig. 5c. As in the case of the SRF-PLL, the PLL system loses track of the grid phase angle and goes into an oscillatory mode. The PLL Counter's response to a similar condition of phase imbalance is shown in Fig. 5d. It is seen that the PLL still maintains a phase error of zero, showing perfect tracking of the grid voltage phase angle.

Figure 6 shows the responses of the different PLL implementations under voltage offset; a dc voltage, which is 10 % of the peak utility voltage is added to phase A at 0.5 s. The response of the SRF-PLL is seen in Fig. 6a. The SRF-PLL outputs an oscillatory phase error, indicating loss of tracking

of the grid voltage under offset conditions. The same oscillatory phase error is observed when the PW PLL system is subjected to an offset grid condition, as can be seen in Fig. 6b. The response of the proposed method is shown in Fig. 6c. The offset is applied at 0.5 s, which causes an initial transient disturbance. However, the phase error is observed to decay to zero in about 0.5 s as is seen in Fig. 6c, after which the system accurately tracks the phase angle of the grid voltage.

The behavior of the different PLL systems under the condition of grid voltage distortion which occasionally occur in three-phase power systems is shown in Fig. 7a. This condition is implemented by the introduction of fifth harmonics among the phases. The response of the SRF-PLL system can be seen in Fig. 7b, where the disturbance introduced at 1 s causes the system to fall out of lock into an oscillatory phase error. A similar phenomenon is seen in Fig. 7c, where the introduction of the voltage distortion at 1 s causes the PW PLL system to lose track of the grid voltage phase angle. Figure 7d shows the response of the PLL Counter when the voltage distortion is introduced to the PLL counter at 0.5 s; we observe the phase error remain unchanged at zero.

5 Experimental Results

5.1 Setup

The proposed PLL algorithm is validated experimentally with a 480 V/60 Hz 3 ϕ AC source fed to a three-phase rectifier. Real-time implementation is achieved using OP4500 platform, a real-time digital simulator with a 4-core 3.3 GHz INTEL processor fully integrated with MATLAB/Simulink from Opal-RT. The experimental setup can be seen in Fig. 8. The parameters used for the three different PLL algorithms and the distortions applied to the grid voltage during the experiment are the same as those used in the simulation and were applied in the RT-LAB environment. The sampling frequency is set to 20 KHz to enable adequate sampling of the input and output signals.

5.2 Results

The behavior of the three different PLL systems under the condition of grid voltage imbalance is observed in Fig. 9. In Fig. 9a, the error response of the synchronous reference frame (SRF-PLL) under phase imbalance can be seen. The voltage of phase C is kept at the nominal value of the utility voltage, while phase B is allowed to sag to a level of 90 % and phase C is allowed to spike to 110 % of the nominal voltage at 4.6 s. It can be observed that at that instant, the phase error stops being zero and starts oscillating instead. The PW PLL is subjected to the same condition of phase imbalance at 5.84 s as can be seen in Fig. 9b. As in the case

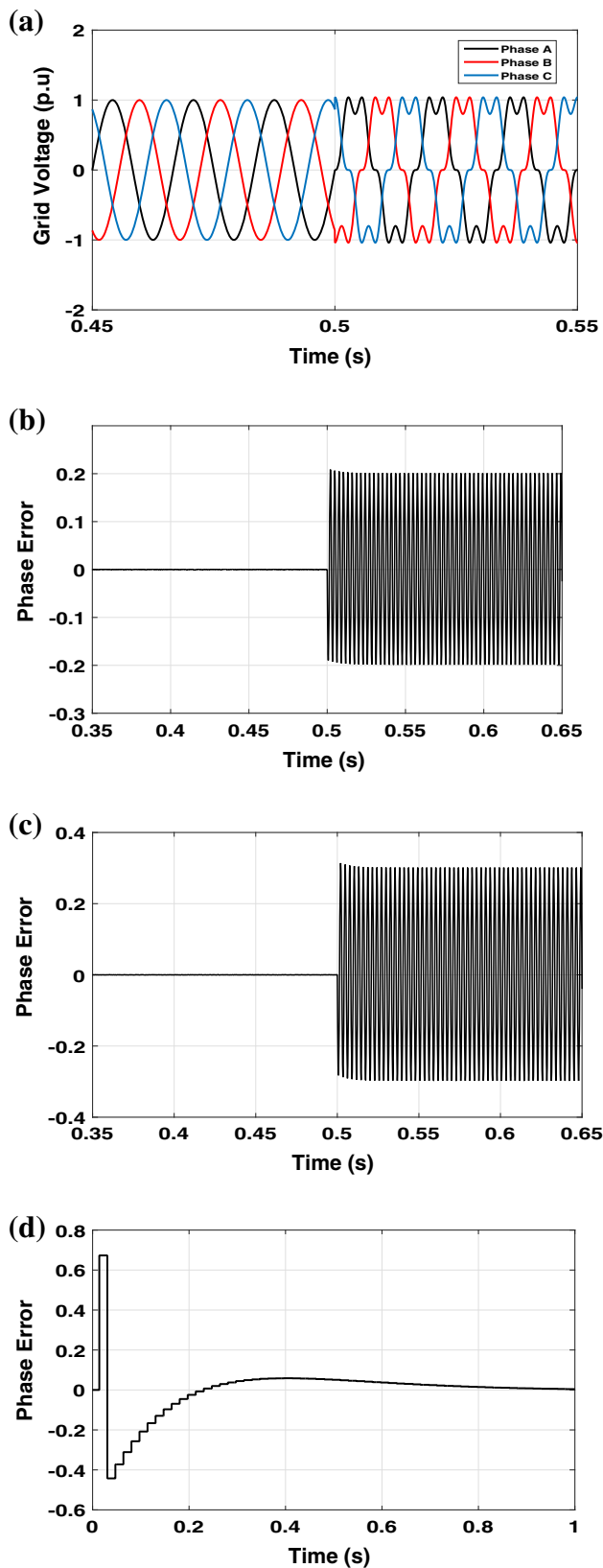


Fig. 7 Response to distorted grid voltage of: **a** grid voltage under distorted conditions. **b** SRF-PLL. **c** PW PLL. **d** PLL Counter

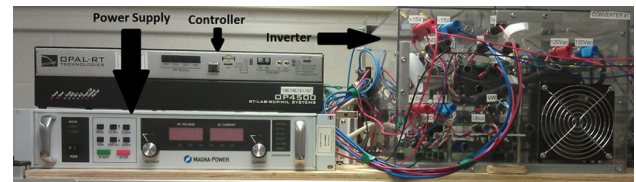


Fig. 8 Setup of the experiment

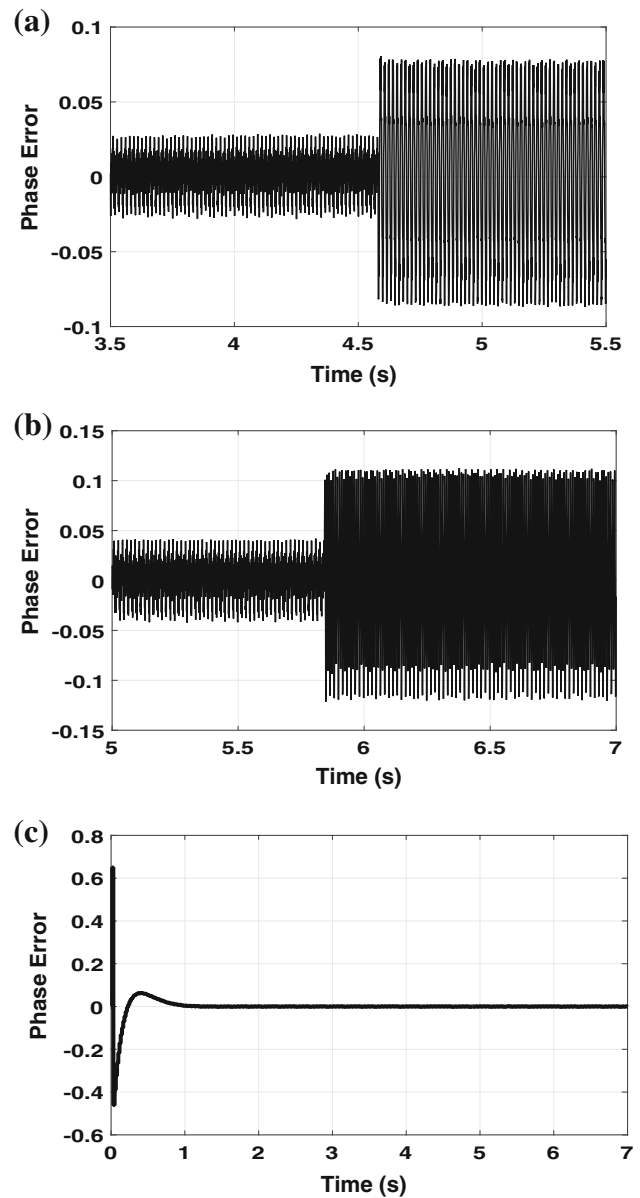


Fig. 9 Experimental system's response to grid voltage imbalance of: **a** SRF-PLL; **b** PW PLL; **c** PLL Counter

of the SRF-PLL, the PLL system loses track of the grid phase angle and goes into an oscillatory mode. The PLL Counter's response to a similar condition of phase imbalance is shown in Fig. 9c where the distortion is applied at 5.28 s. It is seen

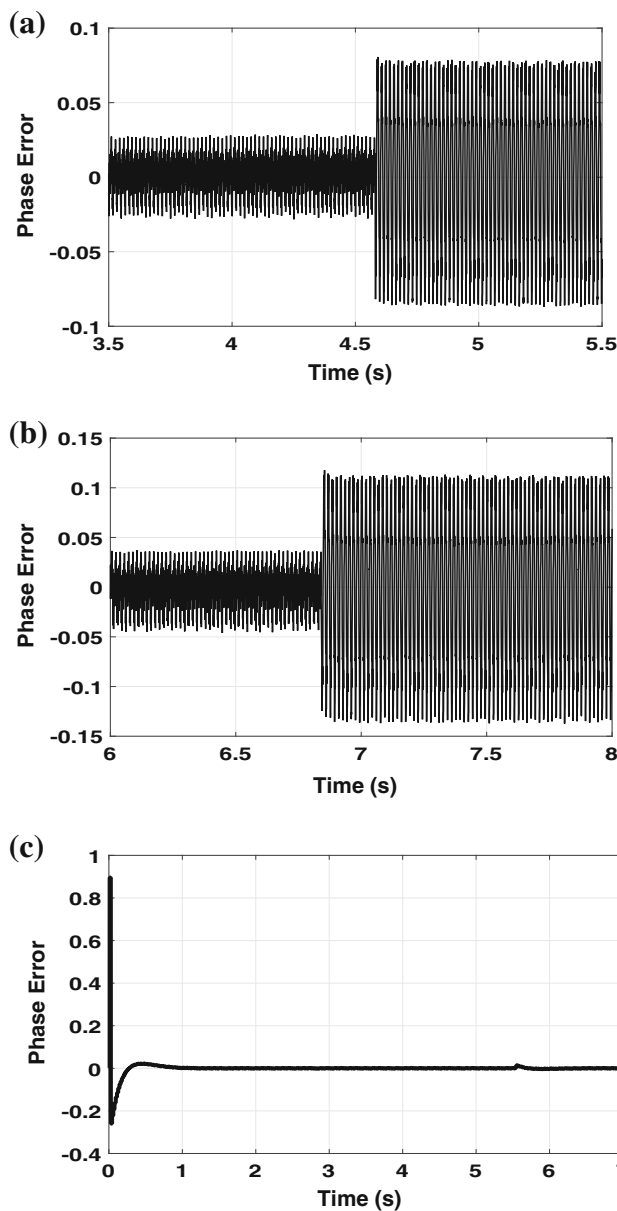


Fig. 10 Experimental system's response to voltage offset of: **a** SRF-PLL; **b** PW PLL; **c** PLL Counter

that the PLL still maintains a phase error of zero, showing perfect tracking of the grid voltage phase angle.

Figure 10 shows the responses of the different PLL implementations under voltage offset. The response of the SRF-PLL is seen in Fig. 10a, where a dc voltage, which is 10 % of the peak utility voltage, is added to phase A at 4.58 s. The SRF-PLL outputs an oscillatory phase error, indicating loss of tracking of the grid voltage under offset conditions. The same oscillatory phase error is observed when the PW PLL system is subjected to offset grid conditions at 6.84 s, as it can be seen in Fig. 10b. The response of the proposed method is shown in Fig. 10c. The offset is applied at 5.55 s, which causes an initial transient disturbance. However, the

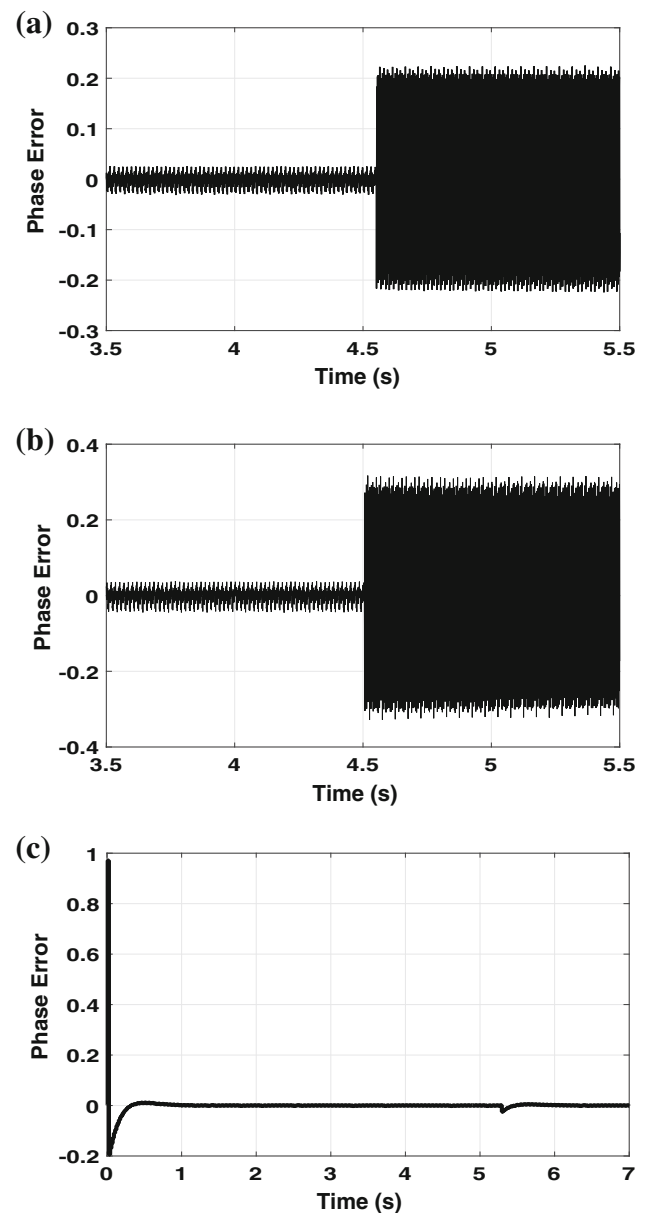


Fig. 11 Experimental system's response to distorted grid voltage of: **a** SRF-PLL; **b** PW PLL; **c** PLL Counter

phase error is observed to decay to zero in less than 0.2 s as is seen in Fig. 10c, after which the system accurately tracks the phase angle of the grid voltage.

Figure 11 shows the response of the different PLL strategies under the condition of grid voltage distortion which occasionally occur in three-phase power systems. This condition is implemented by the introduction of fifth harmonics among the phases. The response of the SRF-PLL system can be seen in Fig. 11a, where the disturbance introduced at 4.55 s causes the system to fall out of lock into an oscillatory phase error. A similar phenomenon is seen in Fig. 11b, where the introduction of the voltage distortion at 4.5 s causes the PW PLL system to lose track of the grid voltage phase angle.

Figure 11c shows the response of the PLL Counter when the voltage distortions are introduced to the PLL counter at 5.29 s; we observe an initial transient in the phase error which decays to zero in about 0.2 s.

6 Conclusion

In this paper, a PLL strategy is proposed for renewable energy systems. The phase detection is implemented using a counter, which gives the system a high level of robustness to distorted utility conditions. The proposed technique achieves robust tracking of the phase angle under grid voltage uncertainties like phase imbalance, distortions and voltage offset. This is demonstrated by its response to the simulations of various grid conditions which were carried out. The three strategies were also validated experimentally using a real-time digital simulator. The performance of the proposed PLL technique is compared to two widely used implementations in the industry: the Synchronous Reference Frame (SRF) PLL and the PW PLL. Simulation and experimental results show effective tracking in the face of practical grid disturbances. The structure of this implementation of this proposed strategy is simple and can easily be realized. The performance of the presented technique is instrumental in achieving the high performance necessary for fully exploiting renewable energies.

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